

BLOCKCHAIN ASSIGNMENT 2

(22B1023, 22B0944, 22B1065)

1. SOME GRAPHS

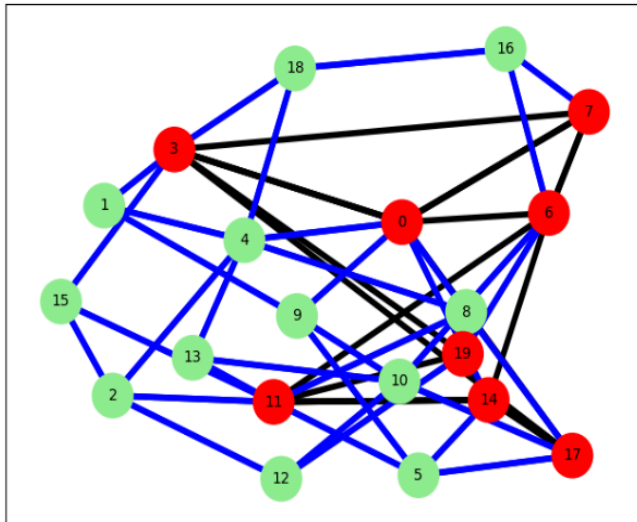


Fig:0 (i) 20 nodes , 40% malicious adjacent graph + attack graph

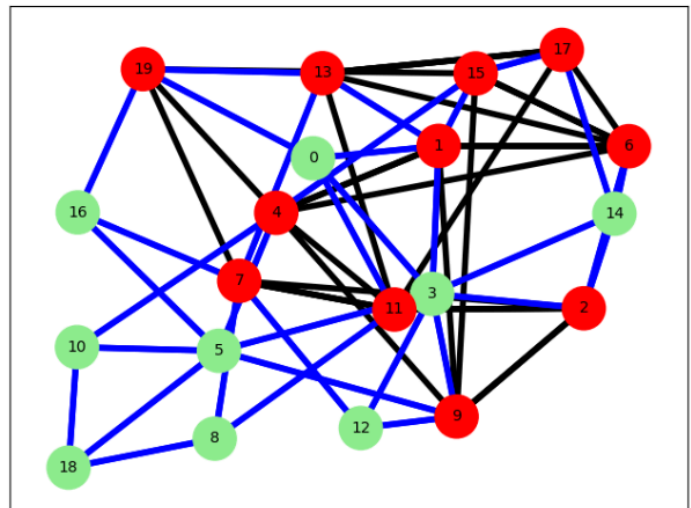


Fig:0 (ii) 20 nodes , 40% malicious adjacent graph + attack graph potential eclipses on node 12 making it vulnerable to eclipse attack thus making it graph completely different in selfish mining + eclipse attack.

2. EFFECTS OF MAICIOUS NODES IN THE LONGEST CHAIN

2.1 Selfish Mining alone

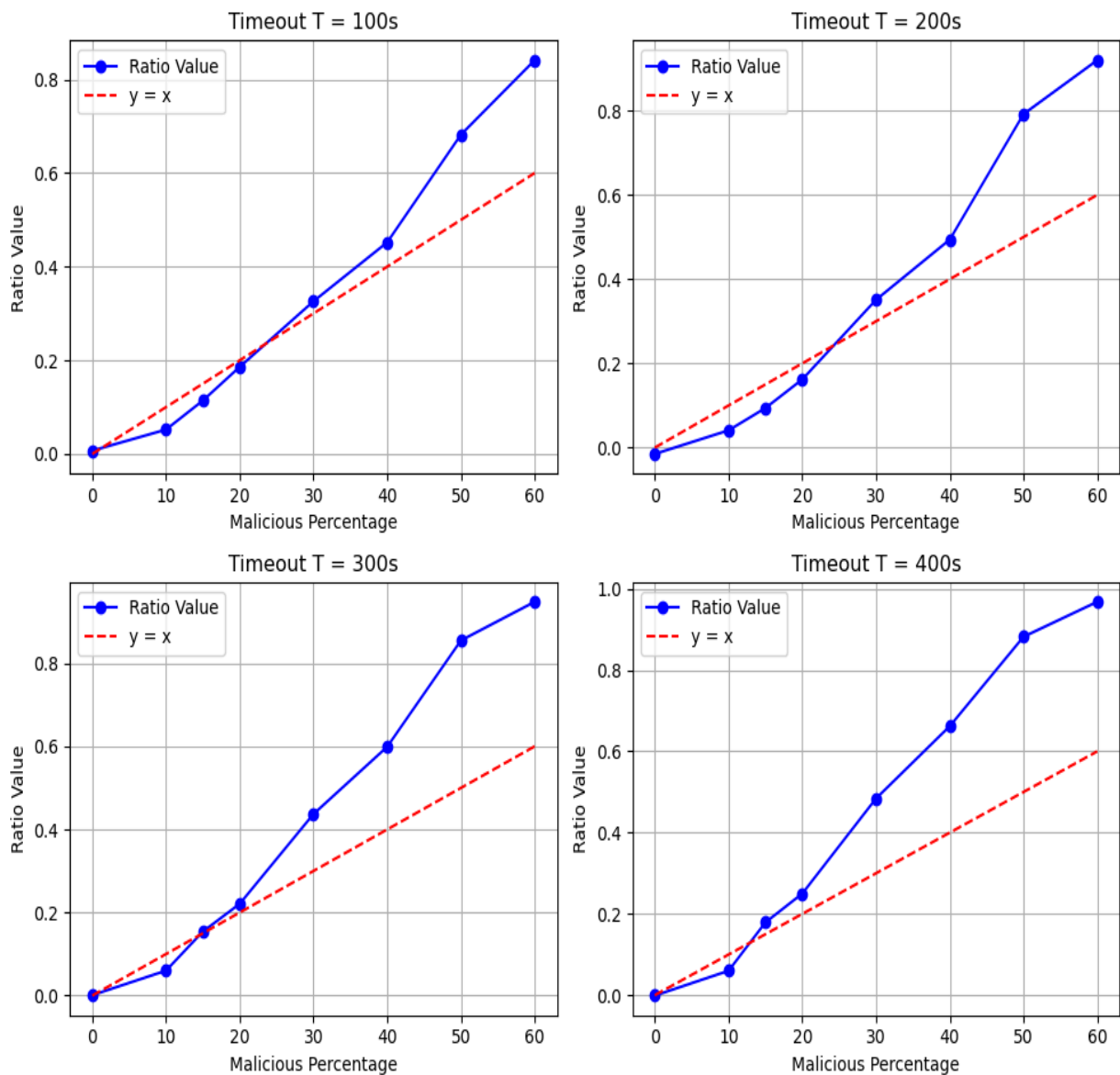


Fig:1 Ratio of malicious nodes' block to total number of blocks in longest chain in ringmasters' block chain shown for different time outs with different percentages for

Malicious nodes for selfish mining only done.

2.2 Selfish Mining + Eclipse Attack

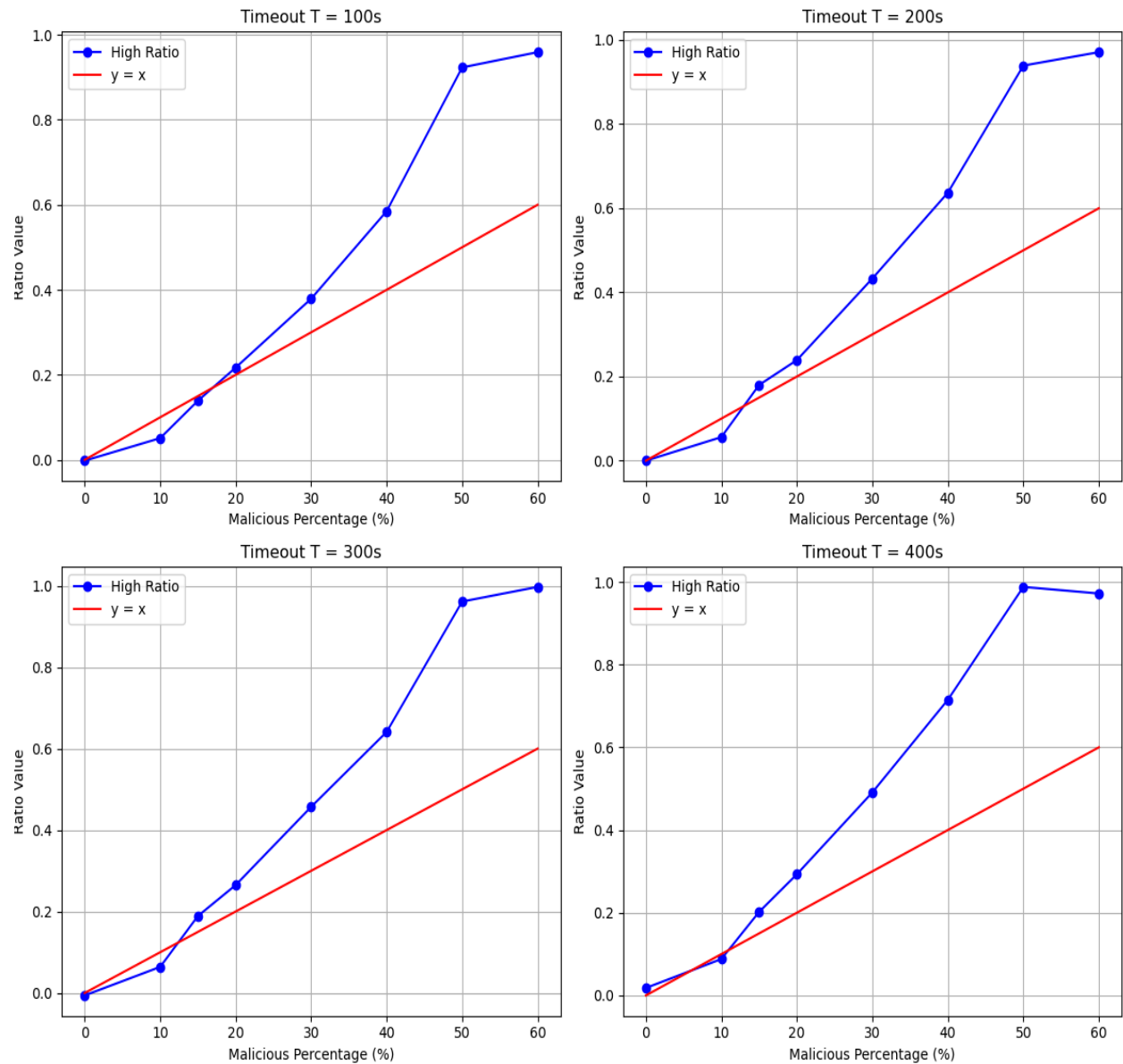


Fig:2 Ratio of malicious nodes' block to total number of blocks in longest chain in ringmasters' block chain shown for different time outs with different percentages for Malicious nodes for the selfish mining + eclipse attack done.

3. EFFECTS OF MALICIOUS NODES IN TOTAL BLOCKS GENERATED

3.1 Selfish Mining Alone

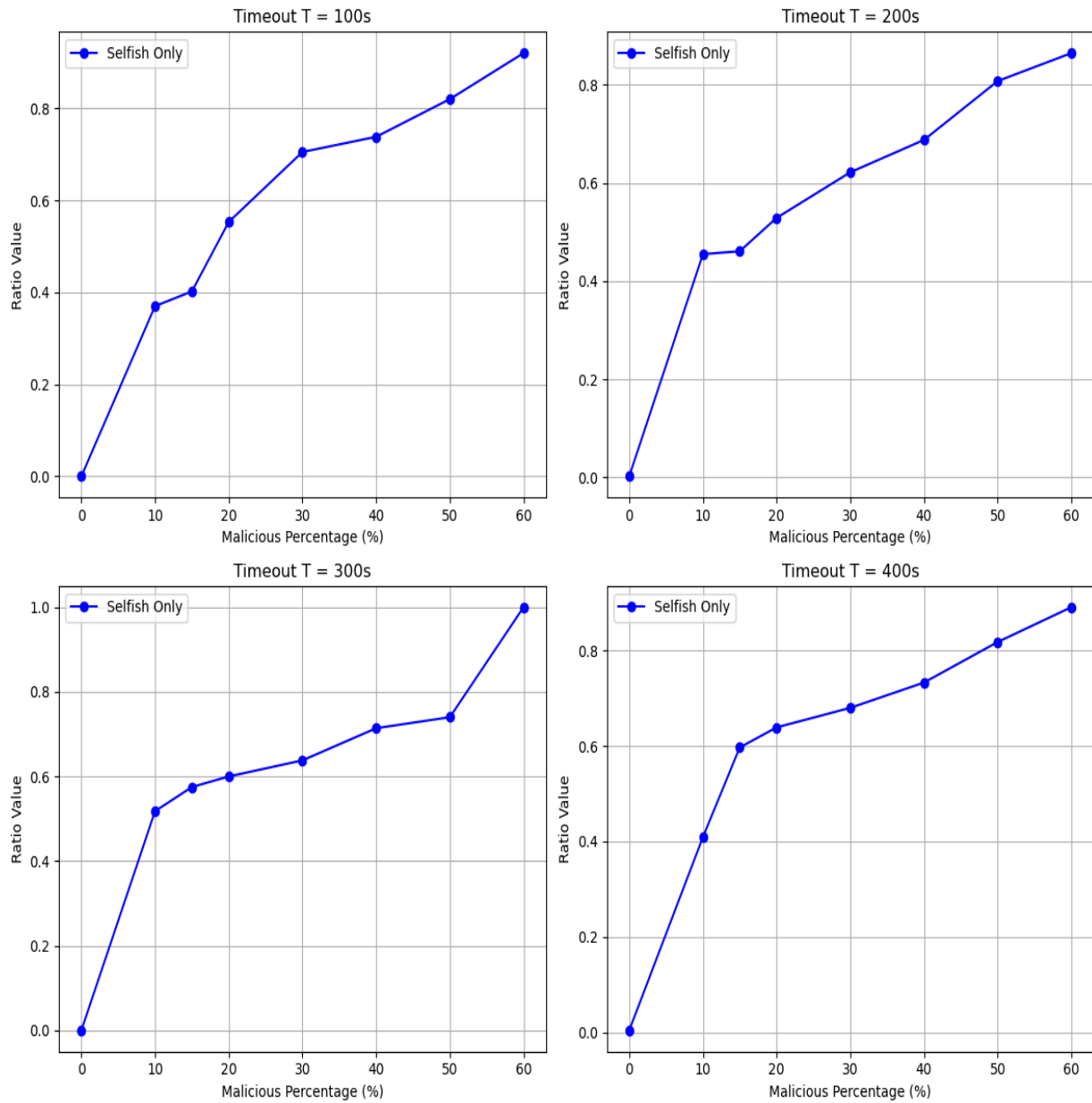


Fig:3 Ratio of malicious nodes' block in longest chain that is present in ringmaster chains to total number of blocks mined by ringmaster i.e malicious nodes mined as a percentage how blocks enter into longest chain of ringmaster for selfish mining done.

3.2 Selfish Mining + Eclipse Attack

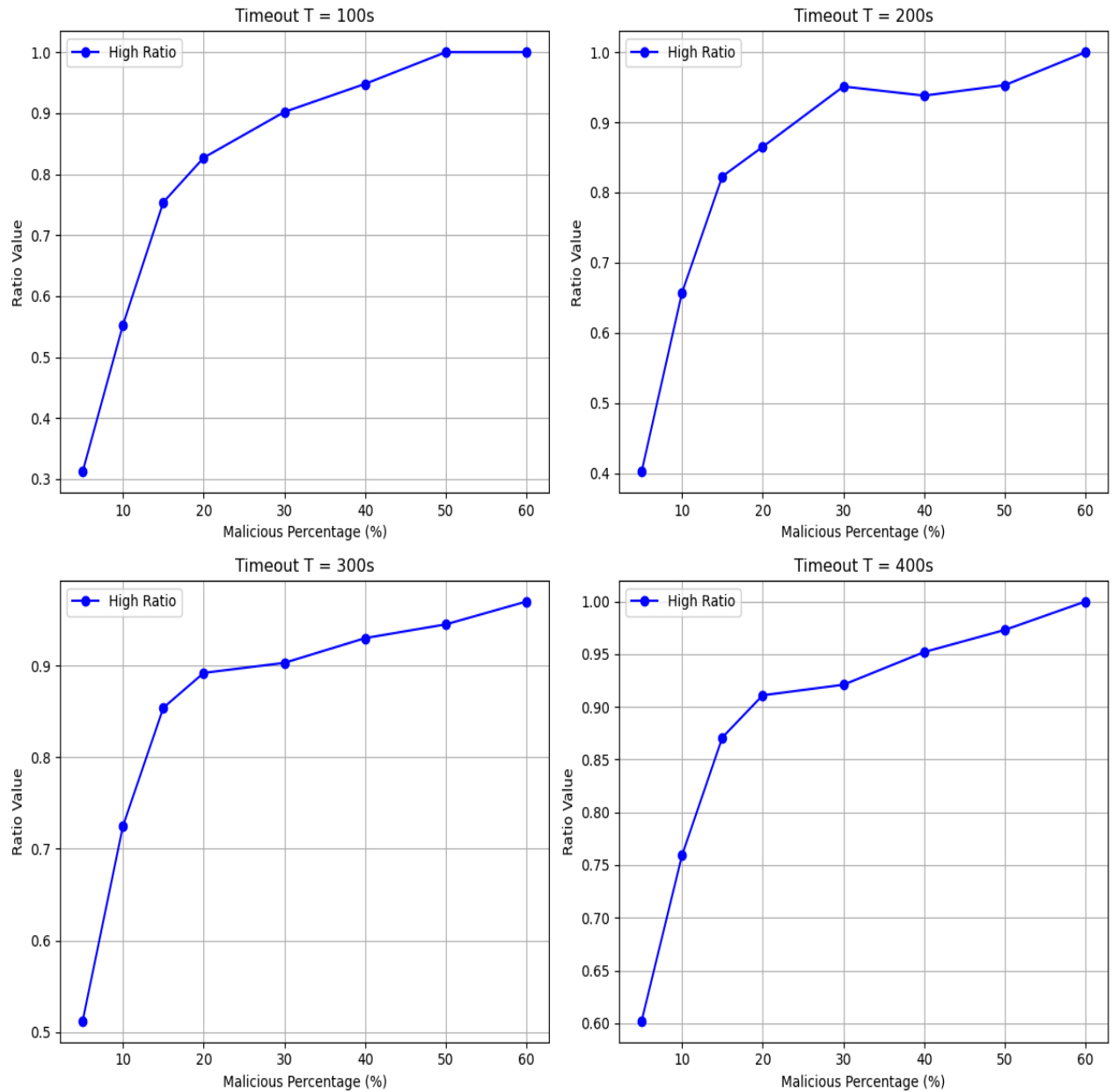


Fig:4 Ratio of malicious nodes' block in longest chain that is present in ringmaster chain to total number of blocks mined by ringmaster i.e malicious nodes mined as a percentage how blocks enter into longest chain of ringmaster for selfish mining + eclipse attack done.

4. Selfish Mining vs Selfish Mining + Eclipse attack (ringmasters' chain)

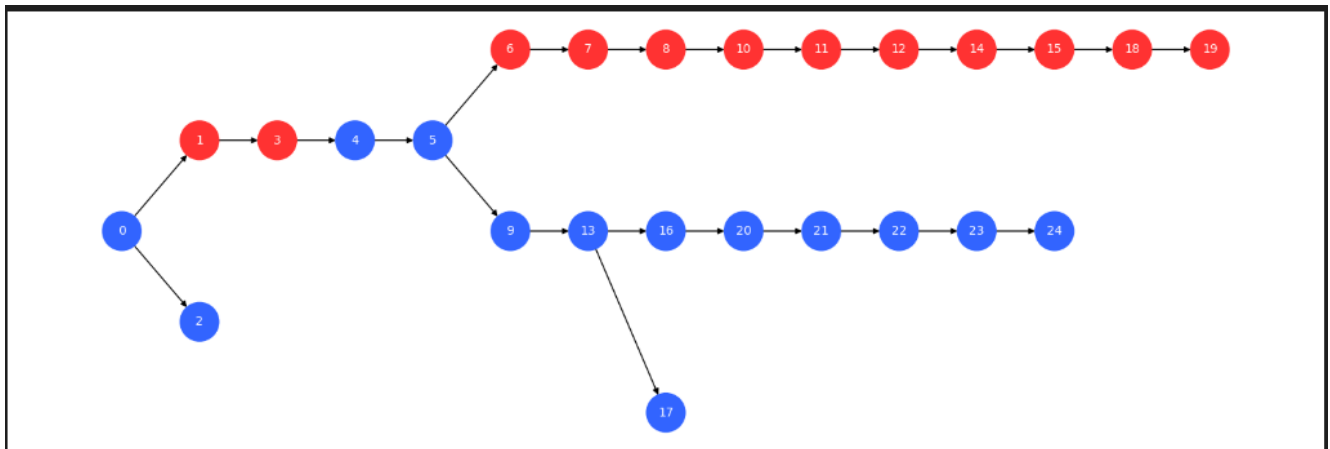


Fig:4 Selfish Mining alone (red is malicious nodes' blocks , blue in honest blocks) for 20 nodes, 40% attack for 200ms timeout

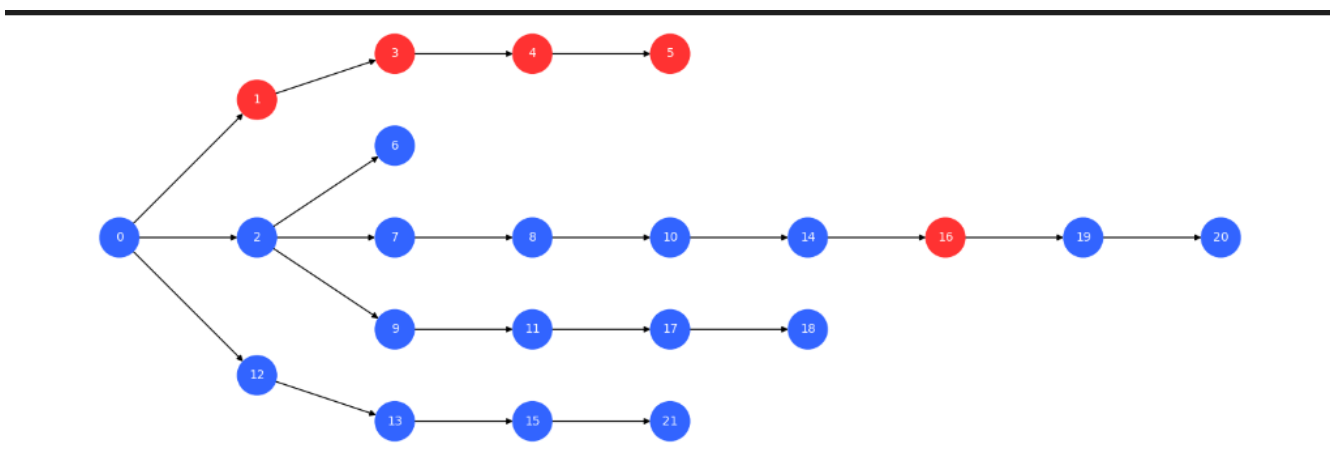


Fig:5 Selfish Mining + eclipse attack done (red is malicious nodes' blocks , blue in honest blocks) for 20 nodes, 40% attack for 200ms timeout

5. Observations and Explanations

- Clearly, as the results indicate, increasing the malicious percentage results in a higher number of malicious blocks in the main chain. This leads to:
 - An increased ratio of malicious blocks in the longest chain compared to the total number of blocks in the longest chain.
 - An increased ratio of malicious blocks in the main chain relative to the total number of malicious blocks mined.

- In Figure 1 and Figure 2, there is an early intersection with $y = x$ at around 0.2 - 0.3. After this point, there is a sharp increase in the ratio of malicious blocks in the main chain, reaching nearly 1 at 0.5. This indicates a 50% attack, where malicious nodes can completely overtake the chain through selfish mining.
 - In Figure 2, the intersection occurs earlier than in Figure 1. This is because, in a selfish mining + eclipse attack, more nodes are mining on their own chain rather than the main chain. This reduces the effective honest mining power, requiring fewer attacker nodes to dominate most of the main chain's blocks. The earlier intersection demonstrates that fewer attackers are needed to launch a successful attack.
 - In Figure 4, the spike is steeper, similar to Figure 3. This happens because, under selfish mining + eclipse attacks, honest nodes are hidden from the network, and the private chain remains concealed. As a result, most blocks are mined in isolated chains, reducing their fraction in the main chain, which is reflected in the ratio graph.
 - In selfish mining alone, the graphs of every node look nearly identical. This is because the private chain and all remaining processes are revealed only after the timeout, ensuring all nodes eventually receive all blocks, keeping the structure the same.
 - Graph disconnection due to malicious nodes. The graph can become disconnected due to malicious nodes not relaying information to the other side of the network. If a group of blocks is fully eclipsed by malicious nodes, only those graphs remain identical, while others differ.
 - In selfish mining + eclipse attacks, we observe more branching compared to selfish mining alone, as malicious nodes isolate specific chains.
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6. Questions

1. Q1: What happens when timeout increases?
 1. Since all honest nodes are isolated into separate groups, there comes a point where eclipsed honest nodes no longer communicate with other honest nodes. The ratio of blocks in a node's chain will contain either isolated honest blocks or malicious blocks.
 2. As timeout increases, the ratio of malicious blocks in the chain increases, since selfish mining releases cause the destruction of the honest chain. Over time, malicious nodes dominate the network as timeout intervals become longer.

2. Q2: What countermeasures can mitigate this issue?

1. Randomized block requesting: Instead of relying on first-come-first-serve, nodes can request blocks from multiple neighbors who claim to have the same hash. By randomly selecting nodes from a maintained set, the probability of obtaining blocks from honest neighbors increases, reducing the effect of the eclipse attack.
2. Tracking block transmissions: Nodes can keep a record of which peers actually send blocks. If a node frequently refuses to send blocks, it can be flagged as potentially malicious. This prevents malicious nodes from withholding blocks and disrupting the chain.

By implementing these countermeasures, the selfish mining + eclipse attack can be weakened significantly.